Morfeas WEB
Help Manual

Current New Web UI

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Introduction

Morfeas WEB is the browser user interface for daily operation and configuration of the Morfeas system. This manual describes the current **new** Morfeas web interface and is intended for operators and service users.

Unless a site-specific deployment says otherwise, open the system in a browser with:

- `http://<device-ip>/`

This manual does not describe the retired legacy web layout.

> **Screenshot Placeholder: Main Landing Page**
> Insert a current screenshot of the ISO Channels Linker landing page, showing the top bar, menu button, search bar, and the main channel table.

Main Page

The landing page is **ISO Channels Linker**. It is the main operational screen.

What You See

The page contains:

- a top bar with the FTP backup directory chip and rolling system summary,
- a search bar,
- action buttons for importing and adding channels,
- the main ISO channel table,
- a right-click context menu for channel actions,
- a menu button that opens all secondary pages.

Main Table

The table shows the configured ISO channels and related runtime information. Typical columns include:

- status,
- ISO channel name,
- device type,
- connection anchor,
- current value,
- history,
- next calibration,
- minimum and maximum values,
- description.

The page refreshes automatically while it is visible, so operators normally do not need to reload manually.

Next Calibration

The **Next Calibration** column is calculated from calibration metadata stored with each ISO channel.

- For **SDAQ** channels, calibration date and period come from the SDAQ runtime calibration data when the device reports it.
- For **IOBOX**, **MTI**, and **NOX** channels, calibration metadata is stored in the OPC UA channel configuration as `CAL_DATE` and `CAL_PERIOD`.
- When a non-SDAQ channel is added and no other calibration metadata is entered, the Add Channel popup defaults `CAL_DATE` to the current date and `CAL_PERIOD` to 12 months.
- The displayed next calibration date is `CAL_DATE + CAL_PERIOD` months.
- A missing or invalid calibration date or period is shown as `--`.

For example, a non-SDAQ channel added on `2026-05-11` with the default 12-month period is shown with next calibration `2027-05-11`.

Measurement Display Rules

The **Meas** column follows the runtime measurement semantics of the current system.

- Normal valid measurements are shown as numeric values with the unit when a unit is available.
- **Out of Range** and **Over Range** continue to show the actual runtime measurement value. They are not converted to an error code.
- Reserved runtime error codes are shown as **red integers** in the **Meas** column.
- Reserved error codes are shown without units and without decimal formatting.
- Reserved error codes are not collapsed to `--`.

Reserved Measurement Error Codes

The following reserved values may appear in the **Meas** column, in OPC UA downstream consumers, and in connected tools that read the same measurement path. These values are emitted by Morfeas core. The web interface displays them when they are present in runtime data; it does not create these numeric error codes from status text alone.

Code	**Meaning**	**Typical Situation**
`-901`	OFF-Line / Disconnected	Handler is alive, but an individual source, receiver, sensor, or channel is disconnected or intentionally unavailable.
`-902`	No Sensor	The device reports that no sensor is present on that measurement channel.
`-903`	Stall / Heating	SDAQ channel is not receiving fresh samples, or NOX is heating / not ready for valid measurement.
`-904`	Unclassified	Runtime data is invalid, but core cannot map it to a more specific condition.

`-905`	Unreachable / Unregistered	Device-level source is gone or unreachable. Examples: handler/source node missing, IOBOX/MTI Modbus transport failure, or retired MDAQ source.
`-906`	Standby / Heater off	Device is reachable but intentionally not measuring. Currently used for NOX heater-off state.
`-907`	Signal Invalid / Data Invalid	Data is arriving but failed validation, usually a telemetry signal-quality or link-layer problem. Currently used by MTI.

Device Status, Web Value, and OPC UA Value

The same reserved value can have a different operator-facing status depending on the device context. Read **Status** first for the reason, then **Meas** for the numeric value sent to downstream consumers.

Web Status below is the linker-table **Status** text. **Web Meas** comes from handler logstat files and usually represents interval data. **OPC UA mapped Value** refers to the linked ISO channel **Value** node used by downstream OPC UA clients.

Device	Web Status / Situation	Web Meas	OPC UA Value	Meaning
SDAQ	Okay	actual value	actual value	Channel is valid and measuring normally.
SDAQ	Out of Range / Over Range	actual value	actual value	Channel reports a real measurement, but it is outside the expected range.
SDAQ	No sensor	`-902`	`-902`	SDAQ reports no sensor on the channel.
SDAQ	Stall	`-903`	`-903`	SDAQ is online, but fresh samples are not arriving.
SDAQ	Unclassified	`-904`	`-904`	SDAQ reports an invalid channel condition without a more specific mapping.
SDAQ	OFF-Line / source missing	`--` when no logstat value is available	`-905` fallback	Linked source node is missing; OPC UA still returns a numeric unavailable value.
IOBOX	Okay	actual value	actual value	Channel is valid and measuring normally.
IOBOX	Unreachable	`-905`	`-905`	IOBOX device-level Modbus transport is unreachable.
IOBOX	Disconnected	`-901`	`-901`	IOBOX handler/device is alive, but the RX/channel path is not responding.
IOBOX	Unconfigured	`--`	`-905` fallback if source node is missing	IOBOX device is reachable, but the linked RX/channel path is not present in runtime logstat. Check IOBOX configuration and wiring.
IOBOX	No sensor	`-902`	`-902`	IOBOX reports no sensor on the channel.

IOBOX	Unclassified	`-904`	`-904`	IOBOX reports an invalid condition without a more specific mapping.
MTI	Okay	actual value	actual value	Channel is valid and measuring normally.
MTI	Unreachable	`-905`	`-905`	MTI device-level Modbus transport is unreachable.
MTI	Disconnected	`-901`	`-901`	MTI handler/device is alive, but telemetry is not responding.
MTI	Radio Disabled / Disabled	`--` when logstat has no channel; otherwise `-901` if core stamped the retained source	`-901` when the retained source node remains; `-905` if source node is gone	MTI telemetry was intentionally disabled or reconfigured. Web does not invent a numeric value.
MTI	No sensor	`-902`	`-902`	MTI telemetry reports no sensor on the channel.
MTI	Signal Invalid	`-907`	`-907`	Packets are arriving, but telemetry data failed validation.
MTI	Error / Unclassified	`-904`	`-904`	MTI telemetry is online, but the data is invalid without a more specific mapping.
NOX	Okay	actual value	actual value	UniNOx sensor is active and measuring normally.
NOX	Handler not running, NOX role removed, or sensor never detected	`--`	`-905` fallback if ISO link exists	NOX source node is missing. Web reports missing logstat/runtime source; OPC UA reports missing mapped source.
NOX	OFF-Line after previously detected sensor timeout	`-901`	`-905` if source node was removed	Logstat records the timed-out sensor as offline; OPC UA may only see the missing source node.
NOX	Heater off	`-906`	`-906`	Sensor is active, but heater-off state makes measurement unavailable.
NOX	Heating / Not in temperature	`-903`	`-903`	Sensor is active, but it is not ready for valid measurement.
NOX	Unclassified / invalid sensor condition	`-904`	`-904`	Sensor is active, but the reported measurement is invalid.

MTI Disabled note: Disabled is an operator-selected radio mode, not a measurement fault. If MTI telemetry is disabled, the web shows **Status** = **Disabled** and leaves **Meas** as `--` unless Morfeas core explicitly provides a numeric channel value. This keeps OPC UA/logstat measurement values owned by core and prevents the web interface from inventing error codes.

NOX mapped-channel note: NOX uses dynamic sensor nodes for UniNOx addresses. If the NOX handler is not running, the bus has been switched away from NOX, or a sensor address has never been detected, the web linker table may show **Status** = **OFF-Line** with **Meas** = — because no logstat runtime source exists. The OPC UA mapped **Value** may return `-905` for the same ISO link because the mapped source node is missing. These two displays describe the same missing-source condition from different layers.

Reading Status Together with Meas

Operators should read the **Status** and **Meas** columns together:

- **Status** = **Out of Range** or **Over Range** with a numeric **Meas** value means the device still reports a measurement, but the channel is flagged as outside the expected calibrated or device range.
- **Status** = **No sensor**, **OFF-Line**, **Disconnected**, **Stall**, or **Unclassified** together with a red negative **Meas** value means the measurement is not valid and the code should be interpreted using the error-code table above.
- **Status** = **Unconfigured** with **Meas** = — means the device is reachable, but the configured runtime channel path is not present.
- **Status** = **Disabled** with **Meas** = — means the function is intentionally turned off, not failed.

Search

Use **Search Input** to filter channels in the current table.

Add Channel Device Search

The Add Channel popup uses device search to choose runtime source paths for new links. Search results are limited to the selected device family and to paths that are not already linked in the current OPC UA channel configuration.

- **SDAQ**: search lists detected SDAQ channels that are not already linked. Single-channel manual offline anchors are allowed when needed.
- **IOBOX**: search lists channels from IOBOX devices whose runtime connection status is **Okay**. A receiver marked **Disconnected** is not listed. Individual channels under an online receiver may still be listed even when their current status is **No sensor** or otherwise invalid, matching the current channel-selection contract.
- **MTI**: search lists channels only when the MTI runtime connection status is **Okay** and the radio/telemetry mode provides channel data. Modes **TC4**, **TC8**, **TC16**, and **QUAD** expose their channel list. **RMSW/MUX** exposes detected **Mini_RMSW** measurement channels. **Disabled** does not expose Add Channel candidates.
- **NOX**: search lists **NOx** and **O2** entries for detected UniNOx sensor objects. The current measurement does not need to be valid; for example, heater-off, warming-up, or previously detected **OFF-Line** sensors may still be linked and controlled later. OFF-Line historical candidates are marked with `(OFF-Line)` in the search result.

The search popup and save validation both enforce device-family matching. IOBOX candidates are not accepted as MTI links, and MTI candidates are not accepted as IOBOX links, even if the physical equipment connected behind the device can be similar.

Context Menu

Right-click a row to access actions such as:

- Edit,
- Delete,
- Scale,

- Calibration,
- Export,
- device-specific actions such as Replace or Replace TC16 when the selected channel supports them.

Keyboard Shortcuts

The following keyboard shortcuts and key actions are currently implemented in the new Morfeas web.

Main ISO Channels Linker

- **Delete**: delete the currently selected channel rows. A confirmation dialog is shown first.
- **Ctrl+Z** or **Cmd+Z**: undo the most recent successful delete. Only one delete step is remembered.
- **Ctrl+Shift+A** or **Cmd+Shift+A**: open the Add Channel popup.
- **Escape**: close the main menu or an open column filter menu.

Add Devices Popup

- **Ctrl+S** or **Cmd+S**: save the device currently being added when the add form is open.
- **Delete**: remove the currently selected removable devices.
- **Escape**: close the popup window.

Add Channel Popup

- **Ctrl+S** or **Cmd+S**: save the current Add Channel form.
- **Enter** in the ISO field: accept the typed ISO code as a custom manual ISO value.
- **Enter** in the sensor path field: accept the typed device path when it matches an available search-pool candidate. Manual offline anchors are intended for SDAQ single-channel use only.
- **Escape**: close the popup window.

Edit Link / Replace Link Popup

- **Ctrl+S** or **Cmd+S**: save the current Edit or Replace form.
- **Escape**: close the popup window.

Calibration Popup

- **Read from SDAQ**: reload the current channel calibration directly from the device.
- **Discard Changes**: restore the last calibration data loaded into the popup.
- **Ctrl+S** or **Cmd+S**: save the current calibration data.
- The popup is fixed to the selected SDAQ channel. No channel reselection is required inside the popup.
- The point table starts with a compact default height and expands automatically when the device reports more used points.

- **Escape**: close the popup window.

SDAQ Scale Popup

- **Ctrl+S** or **Cmd+S**: save the current scale settings.
- The scale popup is separate from the calibration popup and uses its own save workflow.
- The scale form is arranged vertically and uses these field names: **Current Measurement Value**, **Measurement Input Low**, **Measurement Input High**, **Engineering Output Low**, **Engineering Output High**, **Engineering Unit**, and **Scaled Output Value**.
- **Escape**: close the popup window.

Other Popup Pages

- **Escape**: close these popups when they are open: Advanced Settings, Backup & Restore Channels, Check System Status, Network Configuration, Version, and the system control menu.

Menu Overview

The main menu is grouped as follows.

Menu Item	**Purpose**
Network Configuration	Edit hostname and Ethernet IPv4 settings.
Add Devices	View current devices and add or remove supported devices.
Backup & Restore Channels	Configure FTP backup target, create backups, and restore saved bundles.
Check System Status	View live runtime details, log files, and system journal output.
Advanced Settings	View MAC/CAN information, configure NTP server, and manage ISOstandard XML.
System Update	Check for available Morfeas web updates and apply them when desired.
System Reboot	Reboot the device.
System Shutdown	Power off the device.
Version	Show current web and core version information.
Help	Download this manual as PDF.

Network Configuration

Open **Network Configuration** from the menu to edit hostname and Ethernet settings.

Fields

The page contains:

- **Hostname**
- **Mode**: Static or DHCP
- **Network IP**
- **Gateway**

- **DNS addr**

Preset buttons are available at the top of the page for quick loading of predefined values.

Buttons

- **Load Last**: reload the last confirmed configuration into the form.
- **Set**: apply the current values.
- **Close**: close the popup window.

Operator Notes

- If you change the device IP address, reconnect the browser to the new HTTP address after the change is applied.
- DHCP mode disables manual IPv4 entry.
- Presets are convenience values only. Review them before applying.

Screenshot Placeholder: Network Configuration Page Insert a screenshot showing Hostname, Mode, Network IP, Gateway, DNS addr, and the Load Last / Set / Close buttons.

Example Message (Progress):

> Applying configuration: validating and writing system network settings...

Example Message (Success):

> Configuration applied. If the IP changed, reconnect to the new device address in your browser.

Example Message (Failure):

> Apply failed: <error details>

Calibration Popup

Open **Calibration** by right-clicking an SDAQ channel in the main table.

Purpose

The calibration popup is a manual channel calibration editor for the currently selected SDAQ channel. It is used to read, inspect, edit, preview, and save device calibration data.

Page Layout

The popup contains:

- **Upper Section**: device information, calibration metadata, and summary panel share the top area.
- **Current Preview**: `Measured raw value` and `Correct value`.
- **Point Table**: `Uncalibrated value`, `Reference value`, `Offset`, `Gain`, `C2`, and `C3`.

- **Preview Columns**: `Calibrated value` and `Difference (Calibrated - Reference)`.
- **Summary Panel**: calibration type, slope or gain summary, offset, reference minimum and maximum, and unit.

Buttons

- **Read from SDAQ**: reload the calibration directly from the device.
- **Discard Changes**: discard unsaved edits and restore the last data loaded into the popup.
- **Save to SDAQ**: validate the current form and write it to the device.

Preview Behavior

- The **Current Preview** block is a live preview, not a single-row preview.
- `Measured raw value` is read from the current channel raw runtime value when `Raw_Last_Meas` is available in `logstat`.
- `Correct value` is calculated by applying the current full calibration table to the live raw value.
- `Calibration type` is shown only once in the summary panel and is not repeated in every point row.

Validation Rules

- **CH's Unit** must be selected from the supported runtime SDAQ unit list.
- **Used Points** must be an integer between zero and the device maximum.
- Every field of every used point must contain a finite number.
- Empty strings and `nan / -nan` are rejected.
- `Measure` values of used points must be strictly increasing from top to bottom.

Save Semantics

- The calibration popup preserves the entered calibration coefficients as typed after validation.
- It does not automatically recalculate `Offset`, `Gain`, `C2`, or `C3`.
- The separate **Scale** popup has its own dedicated save workflow for SDAQ-I and SDAQ-U.

> **Screenshot Placeholder: Calibration Popup**
> Insert a screenshot of the current calibration popup showing the top shared section, current preview cards, point table, summary panel, and Read / Discard Changes / Save buttons.

Add Devices

Open **Add Devices** to manage the current device inventory.

Devices Table

The upper table shows the current devices with columns such as:

- Bus,

- Device Type,
- Device Name or Address,
- IP Address,
- Status.

The **Status** column shows the current runtime status text reported by the system for that configured device. It is intended as a quick indication of whether the device has been detected and is responding at runtime.

Toolbar

- **Refresh**: reload the current device list.
- **Remove**: remove selected removable devices.
- **Add...**: open the add form.

Add Form

The add form lets the user enter:

- device type,
- CAN interface,
- device name,
- additional device properties required by that device family.

Operator Notes

- Some devices, such as automatically detected SDAQ devices, may not be removable from this page.
- The popup refreshes while active, and pauses when the popup is no longer in focus.
- If the remove action is not available or shows a warning, review whether the selected device is auto-detected and protected from removal.

Channel Replace Workflows

Morfeas WEB supports both single-channel replacement and grouped TC16 replacement.

Single-Channel Replace

Single-channel replace uses the standard Edit Link popup in replace mode. In the current web interface, the standard replace action is offered only for eligible SDAQ channels. IOBOX, MTI, and NOX rows do not show the normal replace option.

Typical Flow

1. In the main ISO channel table, right-click the channel you want to replace.
2. Choose the standard replace action.
3. The popup opens with the source type locked and the target path chosen through device search.

4. Select the replacement device path.
5. Review ISO code, description, min/max, alarms, unit, and calibration fields.
6. Click **Save**.

Type Safety

The replace flow enforces compatibility rules.

> **Screenshot Placeholder: Single-Channel Replace Popup**
> Insert a screenshot of the replace popup showing the locked source type, search button, ISO code, limits, alarms, and Save button.

- Family mismatch is blocked.
- If the source SDAQ subtype is known, subtype mismatch is blocked.
- If the source SDAQ subtype is unknown, replacement is still allowed, but the user is warned to replace within the same type.

Offline Type Display

When an SDAQ channel has been identified before and later goes offline, the UI can display the last known type in a stale form such as:

- `SDAQ-TC1 (last known)`

If the subtype has never been identified, the type may be shown as unknown.

Example Message (Warning):

> Subtype unknown, make sure replace within same type.

Example Message (Blocked Replace):

> Subtype mismatch: expected SDAQ-TC1, got SDAQ-AI8.

TC16 Group Replace

The dedicated **Replace TC16** flow is separate from single-channel replace. It is intended for a full 16-channel SDAQ-TC16 source group.

When the Option Appears

Replace TC16 only appears when the selected source channel belongs to a resolvable offline TC16 group that is complete as channels CH1 through CH16. If the group is partial, the option is not shown and normal single-channel replace must be used.

How It Works

1. Right-click any channel belonging to the full TC16 source group.
2. Choose **Replace TC16**.
3. The UI highlights and selects all 16 source channels.
4. A dedicated Replace TC16 popup opens.

5. The popup shows the source group summary and the source channel mapping.
6. Choose one eligible unlinked TC16 target device.
7. Confirm the operation.
8. All 16 source channels are remapped to the target device in one atomic operation.

Important Notes

- Replace TC16 is all-or-nothing.
- If any validation fails, no channel mapping is changed.
- Target selection is at device-group level, not one channel at a time.
- If no eligible target is listed, there is currently no fully free TC16 target device available.

Screenshot Placeholder: Replace TC16 Popup
Insert a screenshot showing the source group summary, CH1..CH16 mapping chips, available target list, and the Apply Replace TC16 button.

Example Message (Success):

Replace TC16 complete: 16 channels updated.

Example Message (Failure):

Failed to apply TC16 replace. [tc16_target_not_unlinked]

Backup and Restore Channels

Open **Backup & Restore Channels** for FTP-based backup and restore.

FTP Server Setup

The page contains:

- **Host IP**
- **Engine Number**
- **Connect & Fetch**
- **Disconnect**

The default host value is prefilled for the expected FTP server. Credentials are stored server-side and are not edited in the UI.

Backup to FTP

Use **Backup Now** to create and upload a backup bundle for the current configuration.

Restore from FTP

Select a backup file from the list and click **Restore Selected**.

Step-by-Step Example

Connect and Load Backup List

1. Open **Backup & Restore Channels**.
2. Confirm or enter the FTP server IP.
3. Enter the engine number.
4. Click **Connect & Fetch**.
5. Wait for the connection result.
6. If the connection succeeds, the backup list is loaded for that engine directory.

Create a Backup

1. Make sure the FTP connection is active.
2. Click **Backup Now**.
3. Wait for the success message.
4. Confirm that the new file appears in the backup list.

Restore a Backup

1. Make sure the FTP connection is active.
2. Select one file from the backup list.
3. Click **Restore Selected**.
4. Wait for the restore result message.

Operator Notes

- Backup and restore both require a reachable FTP server.
- If the server is unavailable, the page shows an error and no system state is changed.
- The FTP directory chip on the main page shows the currently connected engine directory when configuration exists.
- If two sessions edit the FTP configuration, the page can detect configuration updates from the other session.

Screenshot Placeholder: FTP Backup and Restore Page
Insert a screenshot showing the FTP Server Setup, Backup Now, Restore Selected, and the available backup list.

Example Message (Connect Success):

Connected to FTP at 10.193.135.70. Configuration saved.

Example Message (Connect Failure):

Connection failed: FTP connect failed

Example Message (Backup Success):

Backup complete

Example Message (Restore Failure):

Restore failed: <error details>

Check System Status

Open **Check System Status** to inspect runtime data and logs.

Tabs

The page has three tabs.

Details

This tab shows live system and bus information in tables. Depending on the device and services, values may include:

- CPU temperature,
- uptime,
- bus utilization,
- bus error rate,
- detected SDAQ count,
- incomplete SDAQ count,
- bus voltage,
- bus amperage,
- shunt temperature,
- last calibration timestamps.

Temperature values shown by the current web interface are displayed in degrees Celsius. This includes CPU temperature, SDAQnet shunt temperature, IOBOX/MTI temperature channels, and NOX page shunt temperature.

System Logs

This tab is used for Morfeas application logger files.

- **Preview Logger**: choose one log file to preview.
- **Reload**: reload the logger list.
- **Export Logs**: select one or more log files and export them as one combined text file.

The preview area shows the selected file content in a terminal-style viewer.

System Journal

This tab is used for journal output from system services.

- **Scope**: choose all services, Morfeas system, Apache2, or SSH.
- **Lines**: choose how many lines to load.
- **Reload**: reload the journal view.
- **Export Journal**: export the currently requested journal text.

> **Screenshot Placeholder: System Status Tabs**
> Insert a screenshot showing the Details, System logs, and System Journal tabs with one example content area visible.

How to Use the Tabs

Details Tab

Use this tab first when checking current runtime health. If a service is active but values look stale, compare with the Logs or Journal tabs.

System Logs Tab

Use this tab for Morfeas component logs when you want application-level detail such as handler messages, update logs, backup logs, or runtime component reports.

System Journal Tab

Use this tab when the problem looks system-level rather than application-level, for example:

- service startup failures,
- Apache problems,
- SSH availability,
- reboot or shutdown investigation.

Advanced Settings

Open **Advanced Settings** for machine information, NTP, and ISOstandard management.

Machine Identifier

Displays the MAC address.

SDAQNet (CAN-IFs)

Displays the current CAN interface information for `can0` and `can1`. These values are informational and read-only on this page.

Time Synchronization

The page shows the current NTP server and allows entering a new IPv4 NTP server. Use **Apply** to store the new value.

Example Message (Apply Success):

Applied NTP server: 192.168.1.10

Example Message (Validation Error):

NTP server must be a valid IPv4 address.

ISOstandard

This section lets the user:

- view the currently selected ISOstandard source,
- select a different available ISOstandard file,
- upload a new ISOstandard XML file,
- review the loaded ISO point definitions in a table.

System Update

Open **System Update** to manage Morfeas web software updates.

Current Behavior

The page supports:

- checking whether an update is available,
- applying the update,
- deferring the update,
- reloading the page after service recovery.

Reminder Dot

A red dot on the menu item indicates that the system has recorded an available update. The reminder stays visible until a later check confirms that no update is pending, or until a successful update clears it.

Important Note

The current update flow is intended for the Morfeas web deployment path. Core update is handled separately.

Screenshot Placeholder: System Update Page
Insert a screenshot showing Check Again, Update Now, Update Later, Reload Page, and the status box.

Example Message (Update Available):

Update available. Click "Update Now" to apply, or "Update Later" to postpone.

Example Message (Update Success):

> Update flow completed and web service is reachable. Refresh page (Ctrl+F5) to load latest files.

Example Message (Update Failed):

> System update failed: <error details>

System Reboot and System Shutdown

The system control pages provide single-confirmation actions for reboot and shutdown.

System Reboot

Use **Reboot Now** when you want to restart the device and return to service after boot. The page shows post-action guidance so the operator knows when to reconnect and refresh the browser.

Example Message (Reboot Accepted):

> Reboot command accepted. Wait about 60–90 seconds, then open the device URL again and refresh the browser.

System Shutdown

Use **Shutdown Now** when you want to power off the device. After shutdown, the device must be powered on again manually before the web interface becomes available.

Example Message (Shutdown Accepted):

> Shutdown command accepted. Device will go offline. Power it on manually when needed.

Version

The **Version** page shows version and commit information for:

- the Morfeas web repository,
- the Morfeas core repository,
- the related commit dates.

Use this page when confirming what software is installed on a device.

Troubleshooting

This section gives quick operator guidance for common problems.

> **Screenshot Placeholder: Troubleshooting Section Reference**
> Optionally insert a screenshot collage of the key popups or status messages that operators most commonly use during troubleshooting.

The Web Page Does Not Open

- Confirm that you are using the current device IP address over HTTP.
- If the IP was recently changed, reconnect to the new address.
- If the device was rebooted or updated, wait for services to come back and refresh the browser.
- If still unavailable, check **System Journal** or use SSH to verify Apache and Morfeas services.

FTP Backup Cannot Connect

- Check that the FTP server IP is reachable from the device.
- Confirm that the engine number is correct.
- If the page says configuration was removed, reconnect first.
- If the internal FTP server is currently unavailable, backup and restore cannot be tested end-to-end.

No Backup Files Are Listed

- Verify that the selected engine directory contains backup files.
- Confirm that the FTP configuration belongs to the expected engine number.
- Reload after reconnecting to the FTP server.

Replace or Replace TC16 Option Is Missing

- Confirm that the selected row supports that replace mode.
- For Replace TC16, confirm that the source is a complete offline TC16 group with channels CH1 through CH16.
- If the source group is partial, use normal single-channel replace instead.

No Eligible TC16 Target Is Listed

- No full unlinked TC16 target device is currently available.
- If channels on the target are already linked, the target is blocked.
- Refresh the popup after any device or link changes.

Logs or Journal Look Empty

- Use **Reload** once to refresh the selected tab.
- Verify that the correct logger file or journal scope is selected.
- If the service has not produced messages yet, the view may be temporarily empty.

Red Negative Measurement Codes Are Shown

- `-901` means the handler is alive, but the individual source, receiver, sensor, or channel is offline or disconnected.
- `-902` means no sensor is present.
- `-903` means the channel is stalled, heating, or not ready for valid measurement.

- `-904` means the runtime data is invalid without a more specific mapping.
- `-905` means the device-level source is unreachable or the mapped OPC UA source node is unregistered.
- `-906` means standby / heater off.
- `-907` means signal invalid / data invalid.
- When these codes are shown, the measurement should not be interpreted as a physical live value.

Update Reminder Dot Is Still Visible

- The reminder is based on the stored update flag.
- Run **Check Again** in the System Update page to refresh the latest status.
- After a successful update or a clean re-check, the reminder should clear.

Help PDF Maintenance

The Help menu downloads a PDF file from:

- `docs/help_manual.pdf`

The maintained source for this file is located under:

- `Morfeas_WEB/docs/manual/`

This keeps the active user manual next to the current web code instead of inside the archived legacy documentation tree.

Legacy Documentation Status

The old documentation tree under `LOG-web/Docs/Morfeas_WEB_Docs` is retained only as historical reference. Its structure and API descriptions belong to the legacy implementation and should not be used as the active operator manual for the current web UI.